Shooting while moving for an FTC Decode robot

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Chapter 1

Introduction

In the 2025–26 Decode season of the FIRST Tech Challenge, robots have to shoot "Artifacts" — which are 12.5 centimeter balls — into a goal located 1.4 meters above the ground. While shooting from a stationary position can already prove challenging, some teams wish to save time by shooting while moving.

This article aims to explain the calculations that make this possible, to allow as many teams as possible to implement it. It does not contain any implementation details, but provides the formulas that can be used.

If you want to skip the calculations and go straight to the final result, see 2.4.

Disclaimer: If you are having trouble achieving sufficient precision when shooting from a stationary position, you should probably not consider this approach. Not all physical parameters can be taken into account, which *will* result in a loss of precision. In particular, this text assumes no air friction, because it is difficult to model and has a minimal impact on the trajectories (from what we have observed, at least).

Chapter 2

Calculations

In the following, we use oriented angles. We use (r, θ) for polar coordinates, and $\begin{pmatrix} x \\ y \end{pmatrix}$ for Cartesian coordinates.

2.1 Defining the problem

Consider the following three vectors as projected onto the ground plane:

- $\vec{v}_{robot}$: the velocity of the robot
- $\vec{v}_{shoot}$: the velocity at which the ball is launched relative to the robot
- $\vec{v}_{ball}$: the resulting velocity of the ball in the field's frame of reference

For the sake of simplicity, we assume that $\vec{v}_{robot}$ has an angle of 0. Therefore, $\vec{v}_{shoot}$ has angle α, and $\vec{v}_{ball}$ has angle φ.

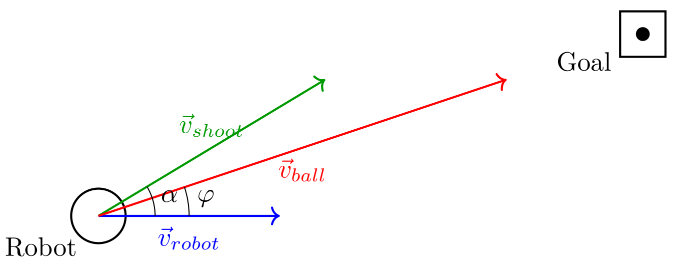

Figure 2.1: *Diagram from above of the moving shot.*

Given $\vec{v}_{robot}$, we wish to calculate $\vec{v}_{shoot}$ such that $\vec{v}_{ball}$ points toward the goal, and has the speed required to reach it.

To do so, we separate the problem into two parts. First, we calculate the required velocity for the ball $\vec{v}_{ball}$, by calculating its ideal trajectory. Then, using basic trigonometry, we can calculate the direction $\alpha = \arg \vec{v}_{shoot}$ and the speed $\|\vec{v}_{shoot}\|$ in which the robot should shoot.

2.2 Calculating the ball speed

This is the most complicated part, as it involves calculating trajectories. To simplify, we will assume no air friction.

2.2.1 Definitions and notations

The following definitions will be essential:

[A] Speed is defined as the derivative of position with respect to time. *Wikipedia*

[B] Acceleration is defined as the derivative of speed with respect to time, which is the second derivative of position with respect to time. *Wikipedia*

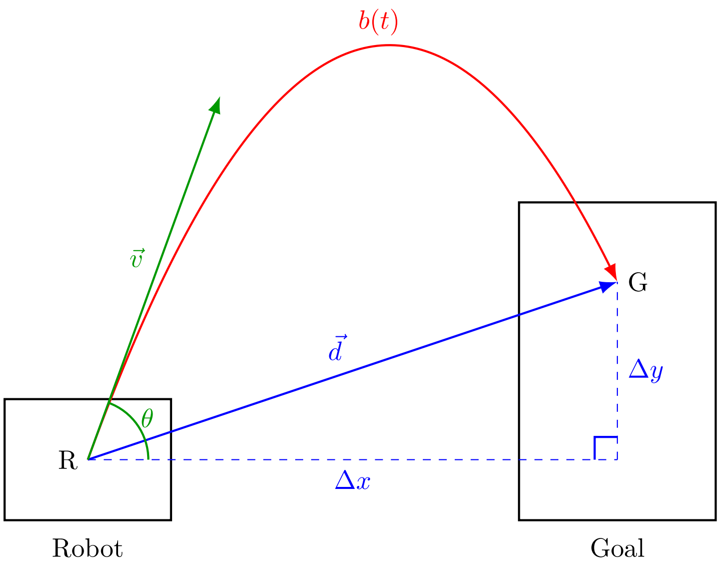

Figure 2.2: *Diagram of the shot. Side view on the plane passing through line RT and perpendicular to the ground.*

Once again, we consider the vectors as projected onto the plane.

Let $\vec{d} = \begin{pmatrix} \Delta x \\ \Delta y \end{pmatrix}$ be the (positive) distance between where the ball stops touching the robot and the point we are aiming for in the goal. Let $\vec{b}(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}$ be the position of the ball with respect to time, and let T be the total duration of the shot, such that $\vec{b}(T) = G$. Finally, let $\vec{v}$ be the velocity of the ball as it exits the shooter, and let $\theta = \arg \vec{v}$ be the vertical angle at which the ball is shot.

To simplify, we make the following assumptions:

- neglect friction so that the ball has a constant horizontal speed
- assume the shooter R is at the center of the robot
- define R as the origin $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

2.2.2 Calculating the trajectory $\vec{b}(t)$

We know the vertical acceleration of the ball is minus the gravitational acceleration g which means that:

$$\begin{cases} y''(t) & = -g \quad \text{(by [B])} \\ y'(t) & = \int y''(t) \cdot dt \end{cases}$$

$$\begin{aligned} \implies y'(t) &= \int -g \cdot dt \\ &= -gt + y'(0) \\ \implies y(t) &= \int (-gt + y'(0))\, dt \\ &= -\frac{1}{2}gt^2 + y'(0)t + y(0) \end{aligned}$$

Since we know that $y(0) = R_y = 0$, we can simply write $y(t) = -\frac{1}{2}gt^2 + y'(0)t$. Basic trigonometry tells us that $\vec{v}_y = \|\vec{v}\| \cdot \sin\theta$, and we know that the speed at $t = 0$, which is $y'(0)$, must be equal to $\vec{v}_y$. Therefore, we can further simplify to $y(t) = -\frac{1}{2}gt^2 + \|\vec{v}\| \sin\theta \cdot t$.

By assumption, we also know that the ball has a constant horizontal speed. Thus, the horizontal speed can be written as the constant $x'(t) = x'(0)$ (by [A]) $\implies x(t) = x'(0)t + x(0)$. Furthermore, we once again know that $\vec{v}_x = \|\vec{v}\| \cdot \cos\theta$ should be equal to $x'(0)$, and that $x(0) = R_x = 0$. We can simply write $x(t) = \|\vec{v}\| \cos\theta \cdot t$.

Combining the two results, we get:

$$\vec{b}(t) = \begin{pmatrix} \|\vec{v}\| \cos\theta \cdot t \\ -\frac{1}{2}gt^2 + \|\vec{v}\| \sin\theta \cdot t \end{pmatrix}$$

2.2.3 Calculating ball velocity $\|\vec{v}\|$

Now that we expressed the position of the ball as a function of the shooting speed $\|\vec{v}\|$, let's solve for $\|\vec{v}\|$. We know that $\vec{b}(T) = G$, so:

$$\begin{aligned} \Delta x &= \|\vec{v}\| \cos\theta \cdot T \\ \implies T &= \frac{\Delta x}{\|\vec{v}\| \cos\theta} \end{aligned}$$

since if $\cos\theta = 0$, your shooter is going to have a hard time scoring anything anyway....

Using $\vec{b}(T) = G$ again, we also have $\Delta y = -\frac{1}{2}gT^2 + \|\vec{v}\| \sin\theta \cdot T \implies \frac{1}{2}gT^2 - \|\vec{v}\| \sin\theta \cdot T + \Delta y = 0$. Recall the quadratic formula, which gives us the following solutions: $T =$

$\frac{\|\vec{v}\|\sin\theta \pm \sqrt{\|\vec{v}\|^2\sin^2\theta - 2g\Delta y}}{g}$. Therefore, we can write:

$$\begin{aligned}
\frac{\Delta x}{\|\vec{v}\|\cos\theta} &= \frac{\|\vec{v}\|\sin\theta \pm \sqrt{\|\vec{v}\|^2\sin^2\theta - 2g\Delta y}}{g} \\
\implies \Delta x \cdot g &= \|\vec{v}\|^2 \sin\theta\cos\theta \pm \|\vec{v}\|\cos\theta\sqrt{\|\vec{v}\|^2\sin^2\theta - 2g\Delta y} \\
\implies \pm\|\vec{v}\|\cos\theta\sqrt{\|\vec{v}\|^2\sin^2\theta - 2g\Delta y} &= \|\vec{v}\|^2 \sin\theta\cos\theta - \Delta x \cdot g \\
\implies \|\vec{v}\|^2\cos^2\theta \cdot (\|\vec{v}\|^2\sin^2\theta - 2g\Delta y) &= \left(\|\vec{v}\|^2 \sin\theta\cos\theta - \Delta x \cdot g\right)^2 \\
\implies \cancel{\|\vec{v}\|^4\cos^2\theta\sin^2\theta} - 2\cancel{g}\Delta y\|\vec{v}\|^2\cos^2\theta &= \cancel{\|\vec{v}\|^4\sin^2\theta\cos^2\theta} - 2\Delta x\cancel{g}\|\vec{v}\|^2\sin\theta\cos\theta + \Delta x^2 g^{\cancel{2}} \\
\implies \|\vec{v}\|^2(2\Delta x\sin\theta\cos\theta - 2\Delta y\cos^2\theta) &= \Delta x^2 g \\
\implies \|\vec{v}\| &= \sqrt{\frac{\Delta x^2 g}{2\Delta x\sin\theta\cos\theta - 2\Delta y\cos^2\theta}} \\
&= \Delta x\sqrt{\frac{g}{2\cos\theta(\Delta x\sin\theta - \Delta y\cos\theta)}}
\end{aligned}$$

2.2.4 Calculating horizontal ball speed $\vec{v}_x = \|\vec{v}_{ball}\|$

We are almost done. We now have $\|\vec{v}\|$, but we are looking to express the horizontal ball speed $\vec{v}_x = \|\vec{v}_{ball}\|$. Thankfully, we know that $\vec{v}_x = \|\vec{v}\|\cos\theta$ by basic trigonometry, so:

$$\begin{aligned}
\vec{v}_x &= \Delta x\sqrt{\frac{g}{2\cos\theta(\Delta x\sin\theta - \Delta y\cos\theta)}} \cdot \cos\theta \\
\implies \vec{v}_x &= \Delta x\sqrt{\frac{g}{\frac{2(\Delta x\sin\theta - \Delta y\cos\theta)}{\cos\theta}}} \\
\implies \vec{v}_x &= \Delta x\sqrt{\frac{g}{2(\Delta x\tan\theta - \Delta y)}} \qquad \left(\text{since } \tan\theta = \frac{\sin\theta}{\cos\theta}\right) \\
\implies & \boxed{\|\vec{v}_{ball}\| = \Delta x\sqrt{\frac{g}{2(\Delta x\tan\theta - \Delta y)}}}
\end{aligned}$$

2.3 Calculating the shooting vector $\vec{v}_{shoot}$

Since the robot is moving with velocity $\vec{v}_{robot}$ relative to the field and launches the ball with velocity $\vec{v}_{shoot}$ in its frame of reference, the resulting velocity of the ball in the field's frame of reference must be $\vec{v}_{ball} = \vec{v}_{robot} + \vec{v}_{shoot}$.

Let r be such that $\vec{v}_{ball} = (r, \varphi)$, and let s be such that $\vec{v}_{robot} = (s, 0) = \begin{pmatrix} s \\ 0 \end{pmatrix}$ (since $\vec{v}_{robot}$ has an angle of 0).

2.3.1 Shooting angle α

We wish to calculate the shooting angle α, such that $\vec{v}_{ball}$ points toward the goal; in other words, we want to find the value of α such that $\vec{v}_{robot} + \vec{v}_{shoot}$ has an angle of φ. We have:

$$
\begin{aligned}
\vec{v}_{shoot} &= \vec{v}_{ball} - \vec{v}_{robot} \\
&= \begin{pmatrix} (\vec{v}_{ball})_x - (\vec{v}_{robot})_x \\ (\vec{v}_{ball})_y - (\vec{v}_{robot})_y \end{pmatrix} \\
&= \begin{pmatrix} r \cdot \cos\varphi - s \cdot \cos 0 \\ r \cdot \sin\varphi - s \cdot \sin 0 \end{pmatrix} \qquad \text{(by basic trigonometry)} \\
&= \begin{pmatrix} r \cdot \cos\varphi - s \\ r \cdot \sin\varphi \end{pmatrix}
\end{aligned}
$$

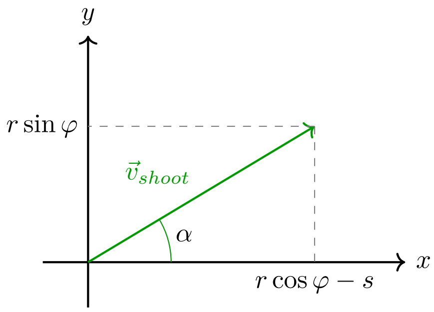

Figure 2.3: *Decomposition of the ball velocity into its horizontal and vertical components.*

Therefore, we are done:

$$
\boxed{\alpha = \arg \vec{v}_{shoot} = \operatorname{atan2}(r \sin\varphi, r \cos\varphi - s)}
$$

2.3.2 Shooting speed $\|\vec{v}_{shoot}\|$

Now that we know the horizontal speed of the ball, we can calculate the horizontal shooting speed $\|\vec{v}_{shoot}\|$. Reusing the previous calculation, we found that $\vec{v}_{shoot} = \begin{pmatrix} r \cdot \cos\varphi - s \\ r \cdot \sin\varphi \end{pmatrix}$. Thus, the Pythagorean theorem gives us:

$$
\begin{aligned}
\|\vec{v}_{shoot}\| &= \sqrt{(r \cdot \cos\varphi - s)^2 + (r \cdot \sin\varphi)^2} \\
&= \sqrt{r^2 \cos^2\varphi - 2rs \cos\varphi + s^2 + r^2 \sin^2\varphi} \\
&\boxed{\|\vec{v}_{shoot}\| = \sqrt{r^2 - 2rs \cos\varphi + s^2}} \qquad \left(\text{by using the identity } \cos^2\varphi + \sin^2\varphi = 1\right)
\end{aligned}
$$

2.4 Final result

Combining the two results gives the final formula for the shooting vector $(norm, angle)$:

$$
\boxed{\vec{v}_{shoot} = \left(\sqrt{r^2 - 2rs \cos\varphi + s^2}, \operatorname{atan2}(r \sin\varphi, r \cos\varphi - s)\right)}
$$

where

$$\begin{cases} r & = \|\vec{v}_{ball}\| = \Delta x \sqrt{\frac{g}{2(\Delta x \tan\theta - \Delta y)}} \\ s & = \|\vec{v}_{robot}\| \\ \varphi & = \text{angle shooter to goal} \\ \theta & = \text{vertical shooting angle} \\ \Delta x & = \text{horizontal distance } \mid \mathit{goal} - \mathit{shooter} \mid \\ \Delta y & = \text{vertical distance } \mid \mathit{goal} - \mathit{shooter} \mid \\ g & = \text{gravitational acceleration } \approx 9.81 \text{ m/s}^2 \\ \vec{v}_{robot} & = \text{shooter } (\approx \text{ robot) velocity} \\ \arg \vec{v}_{robot} & = 0 \quad \text{by assumption (rotate the coordinate system to make it so)} \end{cases}$$

Note: All quantities are expressed in standard SI units (meters, seconds, radians).

If you are not able to choose a shooting speed $\|\vec{v}_{shoot}\|$, and instead use a shooting distance d (for example, by interpolating between a few empirically measured shooting distances), the following formula converts shooting speed to shooting distance. Recall the shoot duration $T = \dfrac{\Delta x}{\vec{v}_x}$ where $\vec{v}_x = \|\vec{v}_{ball}\| = r$, so $d = \|\vec{v}_{shoot}\| \cdot T$ gives us:

$$\boxed{d = \|\vec{v}_{shoot}\| \cdot \frac{\Delta x}{r} = \frac{\Delta x}{r} \cdot \sqrt{r^2 - 2rs\cos\varphi + s^2}}.$$